

SAURABH KASHID

Atlanta, GA

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EDUCATION

Worcester Polytechnic Institute

Masters of Science, Robotics Engineering - [GPA - 3.77]

Jan 2022 – Dec 2023

Worcester, MA

Savitribai Phule Pune University

Bachelors of Engineering, Mechanical Engineering - [CGPA - 8.5]

Aug 2017 – May 2020

Pune, Maharashtra, India

Work Experience

Joulea

Atlanta, GA

Autonomy Engineer

Feb 2023 – Present

- Developed and implemented real-time LiDAR obstacle detection on NVIDIA-Orin for safer drone navigation.
- Coded and integrated GICP algorithm for precise GPS and LiDAR odometry alignment.
- Implemented state machine for drone operation, improving mission control and decision-making processes.
- Investigated and implemented STD loop closure with GTSAM pose graph optimization techniques to enhance lidar SLAM localization accuracy.
- Utilized robot localization for sensor fusion of LiDAR odometry, IMU, and GPS, incorporating a bias estimator from PX4 EKF for height accuracy.
- Created building models from Point Cloud data and defined sensor-aware waypoints for efficient coverage path planning, implementing Patchwork++ for ground segmentation
- Led the integration of MOP Mobile SDK, Onboard SDK, and Payload SDK, guiding contractors in app development, creating C++ structures for OSDK, and performing user acceptance testing and code reviews for effective drone data handling.

PROJECTS

Machine Unlearning with Selective Synaptic Damping (SSD)

Nov 2023 – Dec 2023

- Implemented SSD on ResNet-18 and Vision Transformer models for selective forgetting on CIFAR-10 and CIFAR-100 datasets.
- Achieved improved model accuracy after unlearning, demonstrating the effectiveness of SSD in mitigating the impact of specific training data.

6D Pose Detection using Dense Fusion on Linemode Dataset

Mar 2023 – Apr 2023

- Developed a 6D pose detection system using Dense Fusion on the modified Linemode Dataset.
- Applied state-of-the-art techniques on 6 classes out of 15 of linemode in deep learning and computer vision to improve the performance of the system.

Quad-rotor trajectory tracking

Nov 2022 – Dec 2022

- * Designed Sliding mode controller to track the trajectory of quadrotor.
- * Generated quintic trajectories and tested the controller in gazebo environment.

RRbot Trajectory control

Oct 2022 – Nov 2022

- * Formulated RRbot's dynamic equations using the Newton-Lagrange method and a feedback controller to move it to an equilibrium point.
- * Developed controllers to track cubic trajectories of unknown dynamics, tested in Gazebo.

MAPF for warehouse robots

Nov 2022 – Dec 2022

- * Created a 2D warehouse environment in Python using Matplotlib and applied Conflict-Based Search with Space-Time A* for optimal pathfinding.
- * Compared the performance of Priority-Based, Conflict-Based Search, and Conflict-Based Search with Disjoint strategies.

TECHNICAL SKILLS

Languages and Technologies: Python, C++, Open3d, NetworkX, , Linux, Docker, Pytorch, OpenCV, Tensorflow, MatLab, PCL, ROS, ROS2, Rviz, Gazebo, Git, GitHub, plotjuggler(ROS), GoogleOrtool, SLAM

CERTIFICATIONS

- FAA Part 107 Remote Pilot License

31 Mar 2023